Component Name: Structural composite strut assembly.
Part Designation: STRUT-COMP-X-009
Primary Function: Primary load-bearing structural strut used on both commercial regional airliners and military transport airc
Materials Composition: ["carbon fiber reinforced polymer (CFRP)", "high-modulus PAN-based carbon fiber", "epoxy resin ma
Mechanical Parameters: Dimensions per dual-qualified drawing. Tensile modulus 42 x 10^6 m (high-modulus carbon fiber).
Operational Envelope: Temperature range -54C to +85C. Commercial and military flight profiles. MIL-STD-810 vibration and
Application Interfaces: Mounts to both civil transport fuselage and military cargo aircraft wing structures.
Unique Physical Features: High-modulus carbon fiber composite (1C010 material threshold), dual-platform installation hardw
Intended Platform: Dual use · commercial regional airliners AND military transport aircraft.
Certification Basis: FAA Part 25 civil certification AND military specification qualification.